

Appendix N

Detailed Thematic Analysis of the Comparative Performance Evaluation

N.0.1 Content Completeness and Credibility

This theme captures the expert evaluation of whether the AI-generated literature review is a truly comprehensive and trustworthy representation of the set of papers, maintaining factual accuracy and rigorous scholarly sourcing. This theme is foundational, as a literature review must first be accurate and complete before its organization or style can be judged. Experts evaluated this trustworthiness across four sub-themes: coverage, citation accuracy, citation distribution, and crosschecking.

High-quality coverage in a literature review is defined by the synthesis of major arguments and the inclusion of foundational themes inherent within the set of papers. The feedback suggests that the generated output achieves this by distilling complex information into its most vital form. For example, Expert 13 noted that the system “*captures the essence of the state of the art,*” implying that the tool successfully identifies foundational themes rather than just listing facts.

Structural coherence acts as a secondary layer of coverage, where the “*well chosen*” headings described by Expert 16 serve as a roadmap for the reader. This coherence is not

merely aesthetic, it reflects the logical depth of the synthesis. Expert 19 highlighted this functional alignment, stating: “... *the sections make sense and their [generated] content go according to each section.*” This alignment ensures that the appropriate naming of headings is not a superficial label, but a true reflection of the underlying content.

Experts noted that the efficacy of coverage is significantly diminished when the system introduces **fabricated information**. For instance, Expert 1 explained that he found information that seemed new and interesting, but “*checked the set of papers and found that it was not covered*” which led him to conclude the system had hallucinated. Additionally, **thematic omissions** led to a review that felt skewed or incomplete. Expert 15 noted that the inclusion of certain themes, such as participatory ergonomics, was “*interesting, but ... not relevant to the central theme*” of the provided set of papers.

Furthermore, **structural incongruities** appear to disrupt the cognitive flow for some experts, as seen when the naming of headings “*fails to map accurately*” onto the content. Expert 5 noted that the lack of precision in headings undermines the grouping of themes. This reveals a “*tenuous or contradictory link*” between the heading and its content. Expert 26 provided a detailed critique of this mismatch:

I am supposed to be evaluating the section based on the heading. In this case, the heading is about Theoretical Frameworks and Measures, but the text does not address that. It starts by talking about moral disengagement, which has no direct relationship with the heading of the section.

Imbalances in volume significantly detract from the scholarly utility of a review, creating a tension between brevity and meaningful detail. When there is insufficient depth, the resulting superficial analysis fails to meet the structural expectations of the field. For instance, one expert observed that the draft appeared too “*short for a related work section*” (Expert 3), implying it lacked the granular detail required for comprehensive theoretical grounding.

Conversely, when a review suffers from **excessive breadth**, the scholarly purpose is obscured by “*redundant and unnecessary information*” (Expert 13). This Expert’s verbatim frustration with a text that “*goes into a loop*” highlights how a lack of synthesis creates a “*heavy*” reading experience. This suggests that the failure to “*synthesise [the text] much more*” (Expert 13) is not merely a stylistic flaw, but a barrier to the critical exploration necessary for academic rigour.

The fidelity and validity of the source-to-claim relationship is central to citation accuracy, ensuring that the generated content remains strictly anchored to the provided set of papers. At the highest level of academic integrity, the review demonstrates **citation grounding**. This is evidenced by the expert who found that the citations were authentic and originated directly from the source papers. As Expert 12 noted, “*from what I was able to analyze the citations are correct ... [the tool] does not invent*” any information. This alignment ensures that the foundation of the literature review is trustworthy and empirically-supported.

In contrast, the accuracy in these tools is severely undermined by several forms of citation issues, most notably **citation fabrication**. This occurs when content includes non-verifiable citations, or as Expert 14 put it, references that are “*strange*” because the system “*is inventing them.*” This suggests a vacuum of accountability in the generation process. Furthermore, the integrity of attribution is weakened by **misattribution**. In one instance involving a text on organisational identification, Expert 26 warned that “*some citations are misplaced,*” illustrating how claims are incorrectly linked to the wrong source.

Additionally, the use of **secondary referencing** compromises accuracy by pointing to an intermediate source rather than the original work. This creates a disconnect between the claim and its true foundation:

In the first section it cites a study to identify potential harms where an evaluation has been carried out, but it does not cite the original paper. It is as if it was taking the objective of one of the papers and just copying it, but it does not cite the paper

that says that [the original paper], but instead takes the sentence from the papers I shared with you (Expert 15).

This practice suggests that, while the tool can identify relevant content, it fails to maintain the rigorous chain of custody required for academic integrity. This is because as described in section 4.1.2 LitTraceAI was built under the concept of verifiability and not factuality.

The mechanical placement of citations suggests that the rhythm of a text is a proxy for perceived authority. When citations are distributed in a balance way, the integration of evidence feels strategic rather than performative. Expert 9’s observation of “*well distributed*” citations implies that dispersion creates a narrative where the evidence is woven into the logic of the argument. In contrast, when **citations are concentrated heavily at the end of sentences or paragraphs**, it creates a “*stacking*” effect that disrupts the flow. Expert 16 identifies this as a failure of distribution where “*certain sentences ... have too much*” citations while others are left unsupported, suggesting that an **imbalance in citation density** may lead to a loss of academic credibility.

Finally, **crosschecking** functions as a comparative tool where the AI output facilitates the discovery of relevant research findings. Its primary feature refers to the moment where generated content helps identify new ideas, such as in the topic of Entrepreneurial Orientation and Gender. In this case, Expert 18 noted the positive impact of gender diversity in investment decisions:

Also, this is new: “While management team composition is crucial in investment decisions ... gender diversity within the team can also influence investor perceptions, potentially reducing confidence in the venture’s growth and profit potential.”

The expert was asked a follow-up question about why he did not identify gender diversity in the set of papers. He admitted his focus “*was in Small and Medium Enterprise (SMEs)*”,

which is why he missed the component the tool successfully captured. This interaction illustrates that LitTraceAI does not just reproduce data. It can sometimes capture new research ideas such as gender diversity that the expert had overlooked, thereby providing a positive value-add by expanding the researcher’s intellectual boundaries through automated cross-referencing.

N.0.2 Quality Coherence and Academic Discourse

This theme captures the expert evaluation of the AI output’s ability to achieve a logically sound and scholarly analysis. It moves beyond content verification to assess the execution of the argument itself, examining how the AI structures ideas (logical structure), connects them (logical flow), and performs high-level synthesis (critical analysis).

Logical structure evaluates how well a review sequences research to ensure that information is digested as a coherent whole. High-quality execution requires **thematic clustering** or **chronological sequencing** to delineate the temporal evolution of a field. Expert 2 exemplified this by describing a clear, purposeful progression in the text, noting that *“In general, the content is relevant. It introduces the topic, then contextualises it with examples, and at the end tells me the gaps.”* This suggests that when a deliberate methodological approach is used, the reader can easily discern the hierarchy of ideas. However, when this framework is absent, structural integrity is compromised by **disjointed organization**. This results in a fragmented presentation where the reader struggles to find a logical grouping. For instance, Expert 22 felt the work lacked a cohesive sequencing, describing the experience of reading it as encountering *“a set of clippings, many fragments, but without a solid structure.”* This highlights how the absence of a sustained organizational strategy transforms a literature review into nothing more than a collection of disparate facts.

Logical Flow evaluates the rhetorical cohesion of the literature review. A high-quality literature review is defined by **topic cohesion**, allowing for an effortless reading progression where the argument is maintained within and between paragraphs. Expert 8 highlighted this

success, stating the text was readable from start to finish “*without abrupt changes [because it] has a common thread.*” Conversely, when **topic drift** occurs, the text challenges the reader’s ability to follow the intended argument. This occurs when the story the tool is trying to tell becomes fragmented. Expert 11 elaborated on this lack of guidance:

What happens is that ... I need everything to be aligned, to be telling me a story. But there were, perhaps, a couple of paragraphs in which an idea was cut and another started, and it did not flow well. It does not show that there was a smooth transition ... for the reader.

This breakdown is often exacerbated by **illogical signposting**. Instead of creating true connectivity, the tool may use transitional markers to simulate cohesion. For instance, Expert 27 pointed out that using phrases like “*in contrast*” or “*notably*” in ways that do not “*seem appropriate*” results in “*unnecessary meta-discourse.*” Ultimately, this results in a mimicry of academic writing that lacks the underlying logic required for true narrative continuity.

High-quality critical analysis requires moving beyond simple summarisation to systematically **map relationships**. This is evidenced when the content signals where researchers align or clash, effectively showing the “*studies in favour and against*” (Expert 2). A sophisticated analysis also **identifies research gaps** by highlighting the absence or insufficiency of information. Expert 25 illustrated this by stating:

Its main value lies in the identification of theoretical and regional gaps, by pointing out the scarcity of studies in Latin America and the lack of ICT adoption frameworks adapted to emerging contexts.

However, analytical quality is weakened when content focuses on **individual findings in isolation**. This lack of depth occurs when the text “*does not make comparisons ... [and] fails to argue which studies are similar*” (Expert 12). Such imprecision often results

in the **identification of a research gap that is incomplete** which “*needs more elaboration*” (Expert 7). This failure to synthesise is exacerbated when **the narrative relies disproportionately on a single source**. For instance, in the topic of Value Co-creation in Marketing, Expert 21 noted that while a text may meet basic requirements, it fails if it “*marries itself to the definition of a single author who is Vargas*” without offering a critique or showing the “*tendencies of other authors*” to provide a balanced view.

N.0.3 Stylistic and Technical Rigour

This theme captures the expert evaluation of the AI-generated literature review’s adherence to the mechanical and linguistic standards of academic writing. It focuses on the quality of the language itself, systematically assessing the output’s precision, correctness, and authoritative voice. This rigor was evaluated across three sub-themes: academic language usage, language quality, and narrative tone.

The precision of **academic language usage** is a key indicator of a manuscript’s quality. Success is defined by the **use of correct terminology**, where the content uses scientific language accurately. Expert 3 praised this, noting that the use of terms such as “*human-in-the-loop workflows*” showed “*a good technical level.*” Such precision is supported by providing a **correct definition** for foundational terms, which Expert 1 identified as a positive aspect because a “*clear definition of the concept was presented.*” These elements work together to ensure the reader is provided with comprehensive explanations.

In contrast, technical rigour is compromised when **scientific terms are used in a rudimentary fashion**. This is reflected in the critique that some writing “*does not use academic language*” (Expert 6), signalling a failure to meet the expectations of a scholarly audience. This lack of precision is exacerbated by **unintroduced terminology**, which occurs when specialized jargon is used without background explanation:

However, that explanation is limited to saying that the notion of justice comes from Aristotle, and then it changes the subject. If [the AI] is going to talk about

justice, it must also be explained what is understood by it ... and in the same way with the other perspectives (Expert 26).

By failing to define some key terms, the tool may create a barrier to engagement, making the content appear less authoritative and leaving certain domain-specific terminology underexplained. In the case highlighted by Expert 26, further clarification was important because concepts such as justice can carry different meanings depending on the theoretical perspective being discussed. Without this contextual definition, the explanation may remain unclear.

Language quality evaluates the surface-level **grammatical correctness** of the writing to ensure the reader’s attention remains on the scholarly argument. When a text *“has writing errors”* (Expert 22), it does more than just violate rules; it creates a *“distraction”* that shifts the reader’s focus from the argument to the surface. For instance, the presence of *“strange errors [specifically] in the use of capital letters”* (Expert 22) suggests a failure in the foundational rules of formal composition. These mechanical flaws signal to the reader that the AI lacks the technical precision required for high-quality scholarly output, ultimately degrading the perceived authority of the review.

Finally, **narrative tone** focuses on whether the writing achieves an engaging scholarly presence through an **authentic academic voice**. This is evidenced when experts feel the writing flows naturally. Expert 19, for example, argued that the best results were those *“understood from beginning to end ...”* and felt like a personal creation. This engagement is strengthened by **high readability**, where the tool uses an *“enjoyable style”* (Expert 20) to ensure the review functions as a piece of scholarship rather than a mere summary.

Conversely, narrative quality is stripped of its authoritative persona when it adopts a **robotic tone**. This is characterised by a style that feels automated and stiff. Expert 6 provided a verbatim account of this disconnect:

Let’s see, I mean, I’m going to put it this way, it seems to me like it’s a robot doing things for you like a “boom.” “Boom” is like bullet points, meaning it just lists things. The literature review is not fluid.

This “*boom*” style highlights a lack of natural nuance. Because the machine fails to move beyond a simple list, the resulting presentation remains technically functional but fundamentally artificial, failing to meet the expectations of high-level academic communication.